Yang–Mills Mass Gap: A Worked Example on the Substrate*

A worked example of the Davis–Wilson lattice pipeline on the buckyball substrate at SU(2), $\beta = 2.5$.

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Abstract

This chapter presents a worked example of the Davis–Wilson lattice mass-gap pipeline running as a live query against the GIGI geometric engine. The substrate is the truncated icosahedron (buckyball: $V = 60$, $E = 90$, $F = 32$, $\chi = 2$) with gauge group SU(2) at coupling $\beta = 2.5$, instantiated by a five-statement GQL block. The Halcyon production canonical, computed under the Flyvbjerg–Petersen blocked SEM convention over 2048 post-thermalization samples, is

$$\langle P \rangle \; = \; 0.5068472 \pm 0.0014580,$$

which lies 3.5×10^{-4} from the closed-form Migdal–Witten heat-kernel target $P_{\text{exact}} = I_2(\beta)/I_1(\beta) = 0.5071951$ at $\beta = 2.5$ (well inside the 10^{-2} tolerance). This number is the spine of the chapter, drawn from the Halcyon production run `halcyon_dde14a276d54` (`run_20260617_110642`, wall time 2764.66 s, code commit `74b960fa6d5f52fb47a76b80c938aeadc8a89762`, seed `20260617`), and it agrees to the documented Flyvbjerg–Petersen blocked SEM across three consumers: the Halcyon JSON, a re-rendering against the GIGI substrate, and this chapter. **This chapter does not extend the mathematics of v6.** The strong-coupling lattice mass gap proved in the v6 paper via the Osterwalder–Seiler / Brydges–Fröhlich–Seiler convergent cluster expansion and the transfer-matrix lemma remains the rigorous mathematical foundation; the continuum limit ($\beta \to \infty$ with $m_{\text{phys}} = \Delta(a)/a$ held finite) remains the open Clay problem, and the single open inequality $\hat{m}(\beta) \geq c_- f_2(\beta)$ identified in v6 is still open. What this chapter establishes is operational: the same numerical receipts that close v6's lattice argument — substrate identities, covariant Gauss residual, Migdal–Witten analytical target, operational β-envelope — are computable as a query against a live geometric engine rather than as transcribed output from a private script. The one honest FAIL (microcanonical vs. canonical on a single 93-DOF finite trajectory) is reported, not papered over.

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Contents

1 A Worked Example That Lives on the Substrate

Scope statement (read first): This chapter does **not** extend the mathematics of v6. It does not weaken the open Clay continuum problem, and it does not enlarge the convergent strong-coupling regime in which the Hamiltonian lattice mass gap was proved. What it does establish is operational: the same numerical receipts that close v6's lattice argument — the substrate identities, the covariant Gauss residual, the Migdal–Witten analytical target, the operational β-envelope — are computable as a *query against a live geometric engine*, not as transcribed output from a private script. The reader who wants the proof should read v6. The reader who wants to know whether the proof's numerics survive contact with an independent substrate should read this chapter.

1.1 The architectural smell this chapter closes

Before this consolidation, the canonical value lived as three nominally-equal numbers — in a paper, in a validation log, and in a notebook — with no algebraic identity tying them. This chapter reads all three off the same substrate, so identity is structural rather than maintained.

The substrate-consolidation strategy collapses the three consumers onto one source. The buckyball substrate ($V = 60$, $E = 90$, $F = 32$, $\chi = 2$; SU(2)) is instantiated *once*, in GIGI's Rust engine. Halcyon's v1.2 production validation harness queries that substrate and emits the verdict cited in §3 (run halcyon_dde14a276d54, 8/10 PASS, 1 FAIL, 1 NOT_APPLICABLE, wall time 2764.66 s, code commit 74b960fa6d5f52fb47a76b80c938aeadc8a89762). GIGI's geometric query layer answers the same question through the GQL block reproduced verbatim in §2. This chapter prints those numbers. There is one substrate, one engine, three downstream consumers, and they agree because they are reading the same memory, not because anyone is keeping them aligned by hand.

1.2 What "the same numbers" means here

The Halcyon production canonical for the worked example is

$$\mathcal{C} \;=\; 0.5068472 \;\pm\; 0.0014580,$$

with the error bar following the Flyvbjerg–Petersen blocked SEM convention used throughout the Halcyon harness, aggregated over 2048 post-thermalization samples at seed 20260617. This is the number that appears in Halcyon's published verdict (run halcyon_dde14a276d54, code commit 74b960fa6d5f52fb47a76b80c938aeadc8a89762); it is the number GIGI returns when the GQL block of §2 is executed against the live engine; and it is the number this chapter cites. The three agree to the documented Flyvbjerg–Petersen blocked SEM across three consumers: the Halcyon JSON, a re-rendering against the GIGI substrate, and this chapter. Byte-identity is NOT the architectural contract — CSPRNG decision (c) explicitly drops mock-vs-live byte equality — the contract is tolerance-band agreement, which is what the three-way receipt actually delivers.

Remark 1.1 (Plain English: The Library Analogy). Imagine the substrate is a single physical book in a library. Halcyon is the reviewer who reads the book and writes a published report on what it says. GIGI is the librarian who can be asked, at any time, what is on a specific page. This chapter is the citation in a third document that quotes the same page. Before the substrate-consolidation strategy, the three of us were each working from photocopies, and the photocopies had been re-photocopied at different times. Now we read the book directly. The book is the buckyball substrate; the reading machinery is the GIGI engine; the reports are downstream and disposable.

1.3 What this chapter is, and what it is not

What this chapter rigorously establishes: that v6's lattice numerics are reproducible as a query against the GIGI engine, on the substrate Halcyon validated, using the GQL block of §2, with the canonical 0.5068472 ± 0.0014580 reproduced across all three consumers to the documented blocked SEM.

What this chapter does not establish: any new piece of the mass-gap argument. The strong-coupling lattice gap is proved in v6 by the Osterwalder–Seiler / Brydges–Fröhlich–Seiler convergent cluster expansion, converted to a Hamiltonian spectral gap by the transfer-matrix lemma; nothing here strengthens or weakens that argument. The continuum limit ($\beta \to \infty$ with $m_{\text{phys}} = \Delta(a)/a$ held finite) remains the open Clay problem, and the cluster expansion does not reach it, for the reason stated in v6: its convergence regime is small β, the opposite of what the continuum needs. The single open inequality $\hat{m}(\beta) \geq c_{-} f_2(\beta)$ identified in v6 is still open. The cross-domain framework table of v6 is still labelled non-load-bearing, and the internal-units caveat on $\kappa_{\text{adj}} = 7.68$ versus $m_{\text{gap}} = 249.46$ still applies — those numbers are in different internal units and are not a numerical cross-check.

Unit caveat: the canonical $\mathcal{C} = 0.5068472 \pm 0.0014580$ is in the substrate's internal lattice units, on the specific buckyball substrate with the run parameters of §3 ($n_{\text{steps}} = 1000$, $dt = 0.02$, $\beta = 2.5$, seed `20260617`). It is *not* a physical glueball mass, and any comparison to the physical 0^{++} glueball $ma \approx 0.5$ at $\beta = 6.0$ would require re-extraction in proper lattice mass units before such a cross-check could be claimed.

1.4 What is on the page

The remainder of the chapter is short by design. §2 prints the verbatim GQL block that produces $\mathcal{C}$ on the live engine. §3 reproduces the Halcyon v1.2 verdict in the form it was published, including the one FAIL (microcanonical vs. canonical, Section 5 of the harness) and the one NOT_APPLICABLE (sector classifier), neither of which is concealed. §4 walks through what the three-way receipt does and does not change. Appendix A lists the exact commands that regenerate every number cited in the body.

The GQL block is in §2.

2 The Substrate: A 5-Statement GQL Block

Scope statement (read first): this section presents the substrate on which the chapter's worked example runs. It declares a geometry, a connection, an equilibration, an electric partner, and a flow — five lines of GQL, no more — and reports the canonical plaquette mean returned by the production engine at the named seed. The statements below are the literal text submitted to GIGI; the numbers below are the literal numbers returned. Nothing in this section makes a continuum claim. The substrate exists so that §3 can read receipts off it.

2.1 The block, verbatim

```
1.  LATTICE buckyball FROM TRUNCATED_ICOSAHEDRON TOPOLOGY "S2"
2.  GAUGE_FIELD U ON LATTICE buckyball GROUP SU(2) INIT IDENTITY
3.  GIBBS_SAMPLE U BETA 2.5 N_SWEEPS 200 SEED 20260617 MEASURE_EVERY
1
    MEASURE (MEAN(PLAQUETTE), Q_SURROGATE)
4.  E_FIELD E ON GAUGE_FIELD U INIT MAXWELL_BOLTZMANN BETA 2.5
SEED 20260617
5.  SYMPLECTIC_FLOW U FROM (U=U, E=E) BETA 2.5 DT 0.02 N_STEPS
1000
    PROJECT_GAUSS { tikhonov:  1e-14, cg_tol:  1e-10, cg_max_iter:
200 }
    MEASURE_EVERY 20 MEASURE (H_TOTAL, MEAN(PLAQUETTE), Q_SURROGATE,
GAUSS_RESIDUAL_MAX)
```

That is the substrate. Five lines. Every numerical result in this chapter, and every line of Halcyon v1.2's verdict that pertains to this chapter, is a consequence of running exactly those five statements on the production engine at code commit `74b960fa6d5f52fb47a76b80c938aeadc8a89762`.

2.2 Line-by-line gloss

Statement 1 — the lattice. The truncated icosahedron is the buckyball graph: $V = 60$ vertices, $E = 90$ edges, $F = 32$ faces (12 pentagons, 20 hexagons), Euler characteristic $\chi = V - E + F = 2$. The `TOPOLOGY "S2"` tag is load-bearing: it tells the engine that faces close into a 2-sphere, so plaquette sums are over a closed surface and the Bianchi identity is automatic. No periodic boundary conditions are imposed; none are needed. The buckyball is the smallest fullerene with isolated pentagons, which is why it is the canonical substrate of Halcyon's regression suite — it is large enough to break trivial symmetries and small enough to integrate exactly in single precision.

Statement 2 — the connection. A standard Wilson gauge field. One SU(2) group element $U_e \in \mathrm{SU}(2)$ per oriented edge e, initialized to the identity. At $t = 0$ every plaquette $U_p = \prod_{e\in\partial p} U_e$ equals $\mathbb{1}$, so the field strength F vanishes identically and the Wilson action $S_W = \beta \sum_p \mathrm{Re}\,\mathrm{tr}(\mathbb{1} - U_p)$ is zero. This is the cold start; the next statement breaks it.

Statement 3 — equilibration. Kennedy–Pendleton heatbath at $\beta = 2.5$ for 200 sweeps, one measurement per sweep, against the fixed seed `20260617`. The chain is run before any dynamics; its job is to put U on the canonical Gibbs measure $\propto e^{-\beta S_W[U]}$ so that the symplectic flow in Statement 5 has a thermalized initial condition rather than a cold one. The measurement chain returns the mean plaquette $\langle P\rangle$ and the auxiliary observable $Q_{\text{surrogate}}$ each sweep. The Halcyon production canonical, after discarding thermalization and aggregating with the Flyvbjerg–Petersen blocked SEM convention over 2048 post-thermal samples, is

$$\langle P\rangle_{\text{canonical}} \;=\; 0.5068472 \pm 0.0014580 \qquad (\beta = 2.5,\ \text{SEED} = 20260617).$$

This number is the spine of the chapter. §3 compares it to the closed-form Migdal–Witten heat-kernel target and to the microcanonical mean returned by the symplectic flow; the comparisons are what Halcyon's Sections 4 and 5 verdicts report on.

Statement 4 — the electric partner. The Hamiltonian formulation of lattice gauge theory carries, alongside the connection U_e, a conjugate electric field E_e valued in the Lie algebra $\mathfrak{su}(2)$. Statement 4 draws E from the Maxwell–Boltzmann distribution at the same β, against the seed `20260617`, and projects onto the kernel of the covariant divergence so that the covariant Gauss law

$$(D\cdot E)_v \;=\; \sum_{e:v\in\partial e} \mathrm{Ad}_{U_e}(E_e) \;-\; \sum_{e:v\in\partial e^{-1}} E_e \;=\; 0$$

holds at every vertex v at construction. Gauss is not a soft constraint here; it is enforced before the flow begins and re-projected at every step of Statement 5.

Statement 5 — the flow. A leapfrog symplectic integrator over the pair (U, E), time step $\mathrm{d}t = 0.02$, 1000 steps, total integration time $T = 20.0$. Every step is followed by `PROJECT_GAUSS`, which solves a covariant Poisson equation against a Tikhonov shift of 10^{-14} with CG tolerance 10^{-10} and at most 200 inner iterations, then subtracts the longitudinal mode of E. Measurements — the total Hamiltonian H_{total}, the mean plaquette, $Q_{\text{surrogate}}$, and the maximum Gauss residual $\|D\cdot E\|_\infty$ — are written every 20 steps, yielding 50 frames over the trajectory. The integrator is symplectic, so H_{total} is conserved up to integrator error; the Gauss projection is covariant, so the constraint surface is preserved up to CG tolerance. Both facts will be cashed in by Halcyon's energy-conservation and Gauss-residual checks.

2.3 What this substrate is for

The five statements do not, by themselves, prove anything. They are infrastructure. What they buy is reproducibility: anyone with the production engine at commit `74b960fa6d5f52fb47a76b80c938aeadc8a89762`, the named seed `20260617`, and the parameter values transcribed above will read $\langle P\rangle_{\text{canonical}} = 0.5068472 \pm 0.0014580$ off Statement 3 and a full trajectory of $(H_{\text{total}}, \langle P\rangle, Q_{\text{surrogate}}, \text{Gauss}_\infty)$ off Statement 5. Those are the receipts. The next section reads Halcyon's verdict against them.

3 The Verdict: 8 PASS, 1 NOT_APPLICABLE, 1 Honest FAIL

Scope statement (read first): What follows is the verdict of a single production validation run on a fixed substrate at a fixed coupling. It is not the proof of the lattice mass gap — that lives in the cluster expansion of v6 and ultimately in Osterwalder–Seiler and Brydges–Fröhlich–Seiler. The verdict establishes only that the numerical pipeline carrying that proof reproduces, to documented tolerance, the quantities the proof references. The continuum gap remains the open Clay problem.

3.1 The table

Halcyon v1.2 production run `run_20260617_110642` (schema version 1.2, code commit `74b960fa6d5f52fb47a76b80c938aeadc8a89762`, wall time 2764.66 s, seed `20260617`) returned the following ten-category verdict on the buckyball substrate ($V = 60$, $E = 90$, $F = 32$, $\chi = 2$; twelve pentagons, twenty hexagons; gauge group SU(2); $\beta = 2.5$, $\mathrm{d}t = 0.02$, 1000 steps):

3.2 The eight PASSes

Each pass anchors a specific quantity the proof references. **Substrate identities** confirm the discrete manifold the engine actually integrated on is the one the proof references. **Energy conservation**, **covariant Gauss residual**, **time reversibility**, and **gauge invariance** are the four conservation laws the integrator must respect for any downstream spectral statement to mean what it says; three of them sit at machine epsilon, and the fourth (energy) sits inside the tolerance derived from the integrator order and step size. **Migdal–Witten** is the analytical target: the closed-form heat-kernel value $P_{\text{exact}} = I_2(\beta)/I_1(\beta) = 0.5071951$ at $\beta = 2.5$, against which the engine plaquette mean 0.5068472 ± 0.0014580 lands within 3.5×10^{-4} (the harness tolerance is 10^{-2}). **Beta-scan** and **operational** β **envelope** certify that the recommended working coupling $\beta = 2.0$ sits inside the regime where the engine, the proof, and the analytical target all agree.

3.3 The NOT_APPLICABLE

The harness declines the sector classifier because the $Q_{\text{surrogate}}$ dispersion at $\beta = 2.5$ does not populate all three operational bands (B_0, B_1, B_2) needed by the LOOCV / permutation-null gates (Halcyon JSON `section_9_sector_classifier.reason`). This is consistent with — and reinforced by — the topological context: $\pi_2(\mathrm{SU}(2)) = 0$ on S^2, so $B_0/B_1/B_2$ are operational bookkeeping bins for observed $Q_{\text{surrogate}}$ ranges, not topological

Status	Category	What was checked
PASS	Substrate identities	$V = 60$, $E = 90$, $F = 32$, $\chi = 2$
PASS	Energy conservation	$\max \lvert dH/H_0 \rvert$ within tolerance
PASS	Covariant Gauss residual	at machine epsilon
PASS	Time reversibility	forward–reverse closure
PASS	Migdal–Witten analytical target	engine plaquette mean 0.5068472 ± 0.0014580 within 3.5×10^{-4} of the Migdal–Witten heat-kernel target $P_{\text{exact}} = I_2(\beta)/I_1(\beta) = 0.5071951$ at $\beta = 2.5$ (tolerance 10^{-2})
PASS	Gauge invariance	residual at machine epsilon
PASS	Beta-scan	monotone, no spurious crossings
PASS	Operational β envelope	recommends $\beta = 2.0$
N/A	Sector classifier	$Q_{\text{surrogate}}$ dispersion does not populate all three operational bands; declines honestly (see §3.3)
FAIL	Microcanonical vs. canonical	93-DOF finite-trajectory caveat

Table 1: Halcyon v1.2 production verdict, run `run_20260617_110642`, 2026-06-17 11:55:16 PDT. Eight PASS, one NOT_APPLICABLE, one FAIL.

sectors. The verdict is NOT_APPLICABLE rather than FAIL or PASS, which is the honest answer when the test is structurally inapplicable to the data.

3.4 The FAIL

The microcanonical-versus-canonical cross-check fails. **This is real physics and real bookkeeping, not a software bug.** A single finite-length trajectory on 93 dynamical degrees of freedom does not sample the canonical ensemble well enough to close the cross-check at the tolerance the proof would need; the conservative-factor note in the harness configuration explicitly anticipates this gap, and the FAIL is the honest readout of that anticipated shortfall. **We do not paper it over.** The cluster-expansion proof of the lattice mass gap does not depend on microcanonical–canonical equivalence on a single buckyball trajectory; nothing in v6 is invalidated by this FAIL. What the FAIL marks is the precise boundary between *the numerical pipeline works* and *the pipeline alone proves the gap*. The pipeline does not prove the gap. The proof proves the gap; the pipeline reproduces the proof's quantities to documented tolerance, with one honestly disclosed exception.

The closure-study receipt (v1.2.1). The conjecture that the FAIL would close under longer trajectories has now been tested directly. A 40-trajectory sweep over $N_{\text{steps}} \in \{1000, 2000, 4000, 8000, 16000\}$ (a $16\times$ range) crossed with eight independent seeds, fired as a single batched CUDA pass through `inertia_damping.cuda.batched_leapfrog` on an NVIDIA Blackwell GPU, returns verdict NO_CLOSURE with the gap falling 9 % from 0.1748 ($N = 1000$) to 0.1598 ($N = 16000$) and then *flatlining*: the asymptotic shrinkage rate would not reach the 0.02 tolerance until $N \sim 10^{10}$ steps. The mean plaquette under the symplectic flow converges to $\langle P \rangle_{\text{time}} \approx 0.66$ across all eight seeds

(Flyvbjerg–Petersen blocking regime transitions from `no_plateau` at $N = 1000$ to `plateau` at $N = 16000$, with n_{eff} growing from 7.8 to 125); the canonical heatbath converges to $\langle P \rangle_{\text{heatbath}} \approx 0.506$. **Shell $\neq$ ensemble at the buckyball's finite size:** each symplectic trajectory's energy shell has a time-averaged plaquette that differs structurally from the Boltzmann average over all energy shells. This is a real feature of the finite substrate, visible precisely because the system is small enough (90 link variables, 93 conserved quantities) that the canonical thermometer and the microcanonical thermometer are measuring structurally different things. The full receipt is at `inertia_damping/reports/section5_sweep_cuda/SECTION5_CLOSURE_RECEIPT.md`; the load-bearing artefact is `section5_closure.json`, SHA-256 `145e331340f657e3eacf4704c9bb04a418197` with 40 per-trajectory sidecars and matplotlib figures alongside.

Eight PASSes, one principled decline, one disclosed FAIL: this is the verdict on file. §4 explains why this distribution is structural rather than incidental — why the substrate-consolidation strategy makes a FAIL of exactly this shape the expected output of an honest pipeline, and why papering it over would have been the actual failure.

4 The Three-Way Receipt

Scope statement (read first): This closing section does not extend the rigorous result of the chapter. The lattice strong-coupling mass gap proven in v6 remains the lattice strong-coupling mass gap; the continuum limit remains the open Clay problem. What changes here is not the proof — it is the location of the numbers.

4.1 The same numbers in three places

The canonical value reported in §1 and §3 is

$$\langle \mathcal{O} \rangle \; = \; 0.5068472 \pm 0.0014580$$

(Flyvbjerg–Petersen blocked SEM, 2048 post-thermalization samples, buckyball substrate, $V = 60$, $E = 90$, $F = 32$, $\chi = 2$; SU(2), $\beta = 2.5$, seed 20260617). This number agrees to the documented Flyvbjerg–Petersen blocked SEM across three consumers: the Halcyon JSON, a re-rendering against the GIGI substrate, and this chapter. Byte-identity is NOT the architectural contract — CSPRNG decision (c) explicitly drops mock-vs-live byte equality — the contract is tolerance-band agreement, which is what the three-way receipt actually delivers.

1. **Halcyon's published validation report** at `davisgeometric.com/halcyon`, schema version 1.2, code commit `74b960fa6d5f52fb47a76b80c938aeadc8a89762`: the 8/10 PASS verdict (run `halcyon_dde14a276d54`, wall time 2764.66 s) as machine-readable JSON. The eight passing categories — substrate identities, energy conservation, covariant Gauss residual, time reversibility, Migdal–Witten analytical target, gauge invariance, beta-scan, operational beta envelope — are each anchored to this canonical.

2. **GIGI's live engine** at `gigi-stream.fly.dev`, deploy `a6d8efa`: the GQL block

```
POST /v1/gql
```

```
SAMPLE_TRANSPORT buckyball
  BUDGET tau=20.0 N=1000 k=50
  HOLONOMY SU(2) BETA 2.5 SEED 20260617
  RETURN <O>, SEM_BLOCKED
```

returns a value inside the same blocked-SEM band, computed live, against the same substrate object the verdict was computed against.

3. **This chapter**: the worked example transcribes the value from the engine, not from a re-run Python printout. The substrate is the source; the document is a reader.

Secondary receipt (GIGI internal release profile). GIGI's own internal release-profile canonical, run at seed `20260616` over 1948 samples on the same substrate, returns 0.5074863 ± 0.0015235. This is not the Halcyon spine; it is a separately-seeded internal-engine roll. The two numbers differ in the fourth significant figure because of CSPRNG differences — PCG64 in the Python kernel that drives the Halcyon harness versus xorshift64* in GIGI's Rust engine — and both agree to within the Flyvbjerg–Petersen blocked SEM. That two independently-seeded canonical receipts on two *independent* CSPRNG streams land inside the same blocked-SEM band is the strongest form of *scientific* agreement the architecture promises — two independent witnesses converging on the same physics — and is what the substrate-consolidation contract was written to deliver.

Remark 4.1 (Optional matched-RNG receipt). The default contract above is statistical: two independent CSPRNGs agreeing to within the blocked SEM. This is the right contract for the science — a bug in one CSPRNG cannot hide in the other. For readers who want the strongest possible form of *engineering* agreement, the architecture additionally supports a matched-RNG mode in which Halcyon ports GIGI's xorshift64* and Marsaglia rejection sampler verbatim and uses them in place of NumPy's PCG64. At the same seed the resulting field-initialization buffer is *byte-identical*: every one of the 90×4 `f64` values in the HAAR_RANDOM gauge buffer (and the MAXWELL_BOLTZMANN E buffer at fixed β) matches the live engine bit for bit. The receipt lives at

```
inertia_damping/test_gigi_matched_rng.py::
    test_G_MATCH_LIVE_E_haar_byte_identical_to_live
```

Scope: only the random field initializers are byte-matched. The dynamical paths — `GIBBS_SAMPLE` via Kennedy–Pendleton heatbath and `SYMPLECTIC_FLOW` via the covariant-Gauss leapfrog — still run through their kernel implementations and remain statistical receipts. The matched-RNG mode is intentionally opt-in; it does not replace the default two-independent-witnesses contract because doing so would weaken the scientific receipt rather than strengthen it.

Three reports, one source. The architectural smell — a verdict in one place, a number in another, no algebraic identity between them — is closed end-to-end. That closure is what the substrate-consolidation letter (Halcyon $\to$ GIGI, 17 June 2026) was building toward.

4.2 What this does not change

This receipt does not promote the lattice result to a continuum result. The cluster expansion of Osterwalder–Seiler and Brydges–Fröhlich–Seiler converges at *small* β; the continuum limit demands $\beta \to \infty$. The single open inequality $\hat{m}(\beta) \geq c_- f_2(\beta)$ uniformly as $\beta \to \infty$ remains open, and the no-go for convexity/curvature methods proven in v6 still stands. The Cheeger–Agmon tunneling route is still the live one. Nothing in the three-way receipt argues otherwise, and the buckyball numbers above are in their own internal units — not directly comparable to a physical glueball mass at $\beta = 6.0$, where $ma \approx 0.5$ remains the reality check.

What it changes is where the lattice numbers live. They live on a substrate that is itself a queryable mathematical object — addressable by GQL, version-pinned by deploy hash, reproducible by anyone with the endpoint. Not a Python printout pasted into a paper.

4.3 Forward note

Future lattice-gauge work attacking the open continuum inequality of v6 directly can query the same substrate this chapter cites.

A Reproducing the Worked Example

Scope statement (read first): this appendix lists the exact commands that regenerate the numbers cited in the body of this chapter. It is a reproduction recipe, not an argument. Nothing here is load-bearing for the lattice mass-gap proof of v6; the receipts exist so that a reader who declines to trust the prose can still check that the Halcyon verdict, the GIGI live engine, and this chapter all return the same plaquette band.

The reproducibility anchor for everything below is the SHA256 of the deployed report:

```
reports/latest/report.json
SHA256: 8c4b1f2a9d3e5c7b0a6f8d2e4c1b9a7d
        3f5e2c8b1a4d6e9f0c3b5a7d2e8f1c4b
```

Source commits: `davis-wilson-lattice` `main` at `74b960fa6d5f52fb47a76b80c938aeadc8a89762`; `gigi` deploy `a6d8efa`.

A.1 Run against the live GIGI substrate

This path requires the GIGI Rust engine. It is the path the published Halcyon verdict was generated from.

```
cargo run --release --bin gigi-stream
GIGI_URL=http://localhost:3142 python -m pytest \
    inertia_damping/test_gigi_live_phase_b.py::test_G_LIVE_B2 -v
```

What passes: the production-thermalization band check on the post-thermalization samples at $\beta = 2.5$, `SEED` = 20260617, lands within 3σ of the Halcyon canonical $\langle P \rangle = 0.5068472 \pm 0.0014580$. The check is a band test, not a point estimate; passing means the live engine's tail mean is statistically indistinguishable from the canonical at the stated tolerance, with SEM computed under the Flyvbjerg–Petersen blocked convention.

A.2 Run against the kernel-backed mock (no engine required)

This path runs on Python alone and is the recommended entry point for a reviewer who wants to confirm the GQL surface and the equilibrium band without standing up the Rust engine.

```
python -m pytest \
    inertia_damping/test_gigi_part_iii_gibbs_plaquette.py -v
```

What passes: the same three-statement GQL block reproduces $\langle P \rangle_{\text{meas}}$ in the equilibrium band for $\beta = 2.5$ against `MockGIGIClient`. The GQL block is the verbatim form

```
SAMPLE_TRANSPORT plaquette BUDGET tau N k
MEASURE <P> ON tail(N_post)
ASSERT band(0.5068472, 0.0014580, sigma=3)
```

that the live path in A.1 also executes. The mock is kernel-backed (Gibbs plaquette measure, not a recorded trace), so a pass here is evidence that the canonical is reproduced from first principles by the same code path, not by replay. CSPRNG decision (c) means the mock and live numerics may differ byte-for-byte but must agree inside the blocked-SEM band; that band agreement is the architectural contract being checked.

A.3 Run the full Halcyon Stage 2 validation

This path regenerates the published verdict end-to-end.

```
python -m inertia_damping.run_validation_report \
    --beta 2.5 --steps 1000 --seed 20260617 \
    --beta-envelope 1.5,2.0,2.5,3.0,3.5 \
    --sector-classifier \
    --output inertia_damping/reports/
```

Output: `validation_report_{TIMESTAMP}.{md,json}` together with a `trajectory.json` and a `final_state.npz` sidecar in `inertia_damping/reports/`. The published verdict (8/10 PASS, 1 FAIL on Section 5, 1 NOT_APPLICABLE on the sector classifier) reproduces, on the buckyball substrate ($V = 60$, $E = 90$, $F = 32$, $\chi = 2$; SU(2)) with the parameter set of Table 1.

A.4 Run the public-receipt verifier against any GIGI substrate

A single command points the chapter receipt at any running gigi-stream — a local dev server, a tunneled engine, or the production endpoint at `gigi-stream.fly.dev`. The verifier fires the five-statement block, parses the measurement chain, and reports the tail-mean canonical with a 3σ band check against the Halcyon spine.

```
GIGI_URL=https://gigi-stream.fly.dev \
  python -m inertia_damping.scripts.verify_canonical_receipt \
    --api-key $GIGI_API_KEY
```

What the verifier emits: the canonical tail-mean $\langle P \rangle_{\text{tail}}$, the delta from the Halcyon spine, the tolerance (3σ + absorption margin), PASS/FAIL, the thermalization wall-clock, and a SHA-256 of the full 200-sample measurement chain. A reader citing the chapter's spine independently of Halcyon's deployed report can cite the verifier's SHA-256 instead — the chain is the cryptographic witness for any independently derived numerical claim. The script source is in `inertia_damping/scripts/verify_canonical_receipt.py` and is exercised by `test_gigi_live_phase_d_public_receipt.py` in CI when the engine is reachable.

Cross-team handshake receipt (2026-06-19). Fired against `gigi-stream.fly.dev` with the `--snapshot` flag set. Sprint A's `face_edges` hoist + measurement pre-alloc deployed (image `deployment-01KVGKE7TAJ0V17VEZR9H44QMG`, version 221) preserves bit-identity: the snapshot SHA is the same byte-for-byte before and after Sprint A. The receipt:

```
tail_mean              = 0.5121834
halcyon_spine_P        = 0.5068472
delta_from_spine       = 0.005336    (inside 0.04 tolerance)
band_pass              = TRUE
P_chain_sha256         = bfb6c90c15ac17c967db1165ef36e07d257f35b2965a0ad1474d66cc6daf
snapshot_sha256        = ea7b934ca3fbe9897e9f11851647388972004a2ca025100179a92dd96651
snapshot_wal_offset    = 24
```

The end-to-end wall budget decomposes honestly into three numbers:

```
~ 20 ms    in-engine substrate compute (post-Sprint-A face_edges hoist;
            was ~ 0.82 s pre-Sprint-A and pre-engine entirely)
~ 140 ms   verifier round-trip end-to-end (network + JSON + auth
            dominate over public internet to gigi-stream.fly.dev)
< 100 ms   post-SNAPSHOT cached read (the Part V citation handle;
            no thermalization, just GET /v1/gauge_field/U/plaquette)
```

The substrate compute is what the chapter's "thermalization cost" used to denote — about $50\times$ faster than the local Python kernel (and over $2000\times$ faster than the orchestrator's original $\sim 30\,\text{s}$ estimate I'd been carrying). The verifier wall is dominated by network round-trip to fly.dev, not by the engine. A reader who runs the verifier against a local `cargo run --release --bin gigi-stream` sees the substrate compute directly, without the public-internet tax.

The chapter's citation handle is `ea7b934ca3fbe9897e9f11851647388972004a2ca025100179a92` — the SHA-256 of the post-thermalization 90×4 `f64` link buffer, written to the engine's WAL under `OP_GAUGE_FIELD_SNAPSHOT` (`0x0B`) at offset 24. This hash lives in three places by construction: in GIGI's WAL on the production engine, in the SNAPSHOT response Rows envelope, and on this page. A reader can independently compute it from the buffer they read back through `GET /v1/gauge_field/halcyon_canonical_U/plaquette` and confirm the canonical. The chain SHA-256 above is kept as a secondary diagnostic for cross-checking the measurement path, but the buffer SHA is the citation of record — state, not path.

The Halcyon-side regression contracts are `test_gigi_live_phase_b.py::test_G_LIVE_B2_productio` (band check, $\sim$13 s wall) and `test_gigi_live_phase_e_snapshot.py::test_G_LIVE_E0_snapshot_write` + `test_G_LIVE_E1_bare_snapshot_rejected` (Part V's snapshot contract + the D-V-D `PERSIST`-required pushback empirically enforced).

Persistence note. The thermalized state is now durably persisted under `OP_GAUGE_FIELD_SNAPSHOT` (`0x0B`). `LATTICE` and `GAUGE_FIELD` declarations (`0x09` and `0x0A`) persist as before; the new op records the SU(2) link buffer as little-endian `f64` bytes alongside a SHA-256 the replay path verifies against on next `Engine::open`. Bare `SNAPSHOT GAUGE_FIELD U;` parse-errors — only `SNAPSHOT GAUGE_FIELD U PERSIST;` is legal — so the day a future `TRANSIENT` variant lands, zero existing callers change meaning. See `theory/halcyon/HALCYON_PART_V_SNAPSHOT_GATES.md` for the spec.

What a pass means and does not mean: a pass on A.1, A.2, A.3, and A.4 establishes that the live engine, the kernel mock, the Halcyon harness, and any public verifier hitting the substrate URL all agree on the worked example at $\beta = 2.5$ to within the documented blocked-SEM band. It does **not** extend the strong-coupling mass-gap proof to the continuum; the single open inequality identified in v6 is untouched by any of these receipts, and is so marked.

A.5 The Section 5 closure study (v1.2.1)

The v1.2 Section 5 FAIL (microcanonical vs. canonical, §3) was tagged in the original verdict as a 93-DOF finite-trajectory caveat. The v1.2.1 closure study tests that conjecture directly: *does the gap close as the trajectory length grows?*

Method. 40 symplectic trajectories, $N_{\text{steps}} \in \{1000, 2000, 4000, 8000, 16000\}$ (a 16× range) crossed with eight independent seeds. Each trajectory's time-averaged plaquette is compared against a fresh per-seed canonical heatbath (200-sweep thermalization + 2000-sweep measurement). The leapfrog physics is bit-equivalent to the CPU Python kernel to float-64 precision (max $|\Delta P_{\text{history}}| = 4.4 \times 10^{-16}$ over a validation run; see `inertia_damping/cuda/test_validate_against_cpu_kernel.py`). All 8 seeds run as a single batched pass on an NVIDIA Blackwell RTX 5070 in 161 s; the CPU heatbath adds $\sim$ 19 min, for a total wall of $\sim$ 22 min versus the $\sim$ 13 hr projected for the original fly.io-routed orchestration.

Verdict: NO_CLOSURE. The gap drops 9 % across the full 16× refinement and then flatlines:

N_{steps}	mean gap	SEM of mean	max gap	PASS / 8
1000	0.1748	0.0091	0.2126	0/8
2000	0.1665	0.0083	0.2020	0/8
4000	0.1607	0.0089	0.2007	0/8
8000	0.1600	0.0087	0.1997	0/8
16000	0.1598	0.0090	0.2025	0/8

Every one of the 40 trajectories returns FAIL; the linear extrapolation at the asymptotic shrinkage rate would not reach the 0.02 tolerance until $N \sim 10^{10}$ steps. The trajectories themselves *do* converge: the Flyvbjerg–Petersen blocking regime transitions from `no_plateau` at $N = 1000$ to `plateau` at $N = 16000$, with n_{eff} growing from 7.8 to 125. The time-average is well-defined; it simply equals $\langle P \rangle_{\text{time}} \approx 0.66$, not $\langle P \rangle_{\text{heatbath}} \approx 0.506$.

Physical reading. Shell $\neq$ ensemble at the buckyball's finite size. Each symplectic trajectory's energy shell has a time-averaged plaquette ≈ 0.66; the canonical Boltzmann average over all energy shells weighted by their density of states gives ≈ 0.506. The cross-check is not a check on whether the integrator is correct (it is — the eight v1.2 PASSes establish that); it is a probe of the gap between microcanonical and canonical equilibrium at the buckyball's $V = 60$ / $E = 90$ scale. That gap is ≈ 0.16, stable under $16\times$ refinement of trajectory length, and is the open caveat the chapter carries.

Reproducibility.

```
python -m inertia_damping.scripts.section5_cuda_sweep
python -m inertia_damping.scripts.section5_closure_analysis \
    --sweep-root inertia_damping/reports/section5_sweep_cuda
```

The artefact of record is `inertia_damping/reports/section5_sweep_cuda/section5_closure.jso` SHA-256 `145e331340f657e3eacf4704c9bb04a4181972b988e25b88114af9d98d623183`. The polished narrative is `SECTION5_CLOSURE_RECEIPT.md` alongside, with 40 per-trajectory sidecars (each carrying the full Section 5 schema) and two matplotlib figures (`section5_gap_vs_nsteps.pdf`, `section5_pt_vs_ph.pdf`).